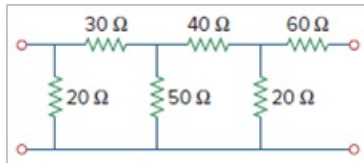


Problem

Obtain the **ABCD** representation of the circuit in Fig. 19.48.

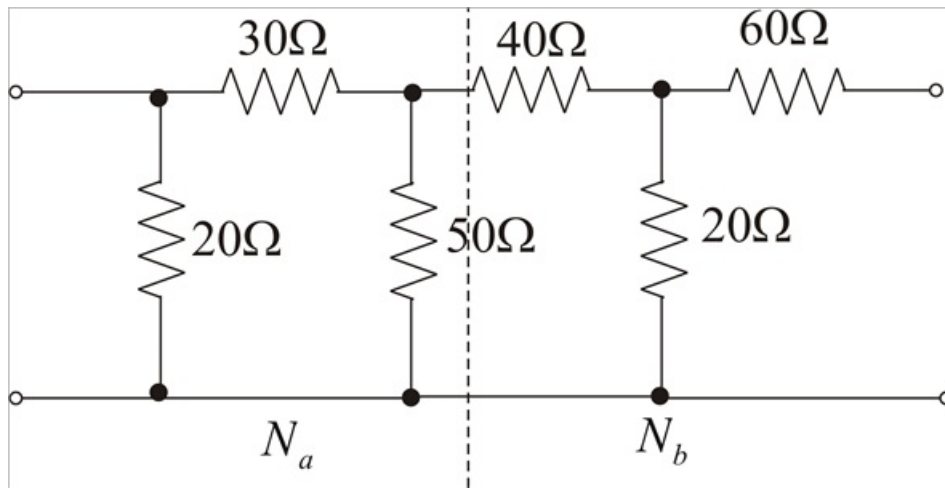
Figure 19.48



Step-by-step solution

Step 1 of 10

Given circuit is



Step 2 of 10

We can consider the given circuit as a cascade connection π and T networks. Thus, we separately show the π and T networks has the transmission parameters.

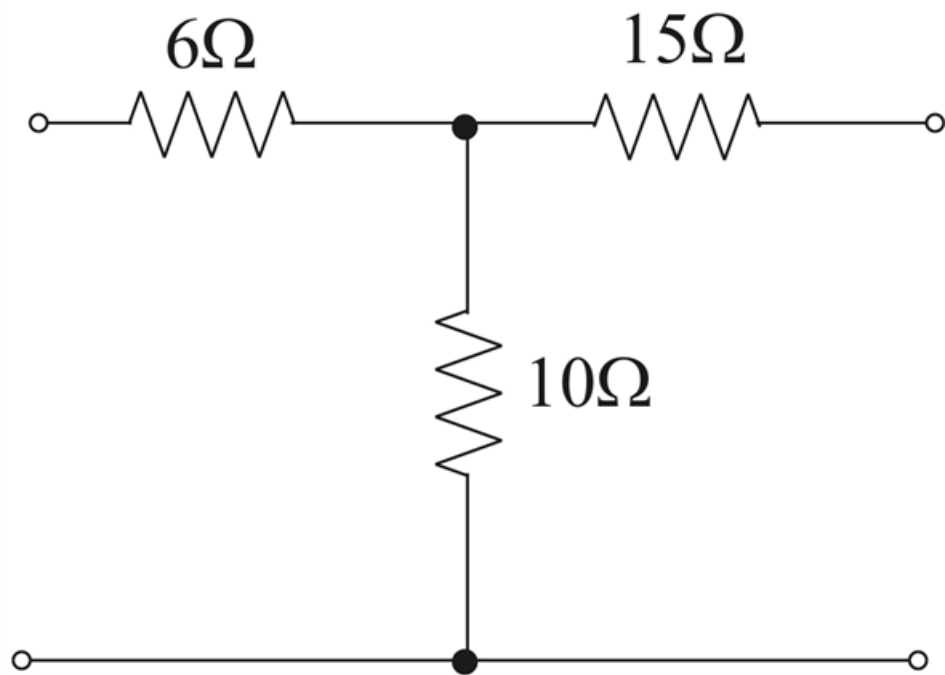
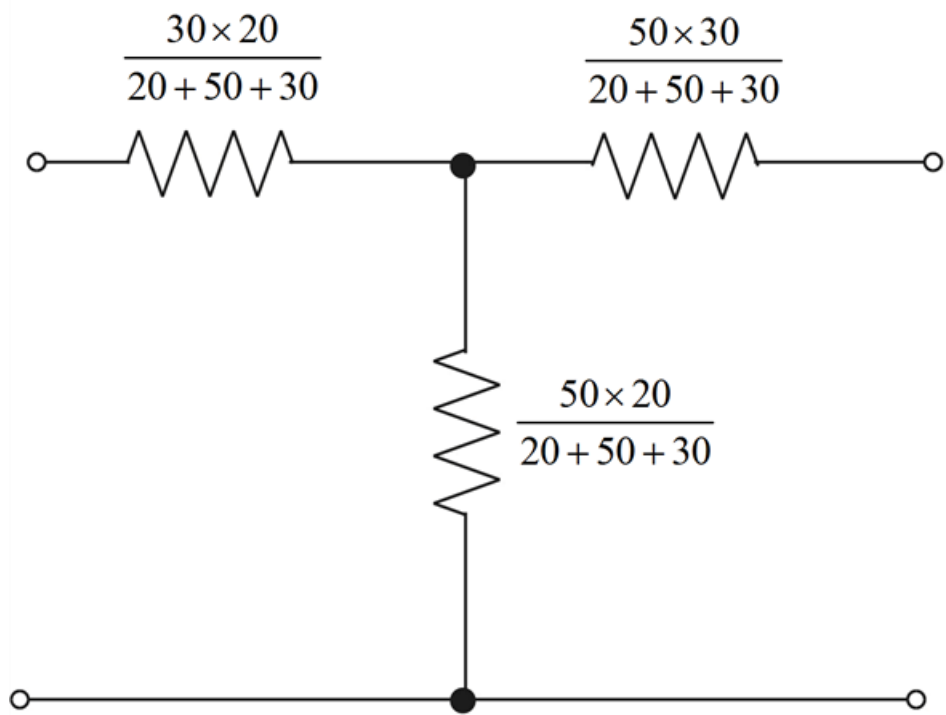
The transmission parameters are

$$V_1 = AV_2 - BI_2$$

$$I_1 = CV_2 - DI_2$$

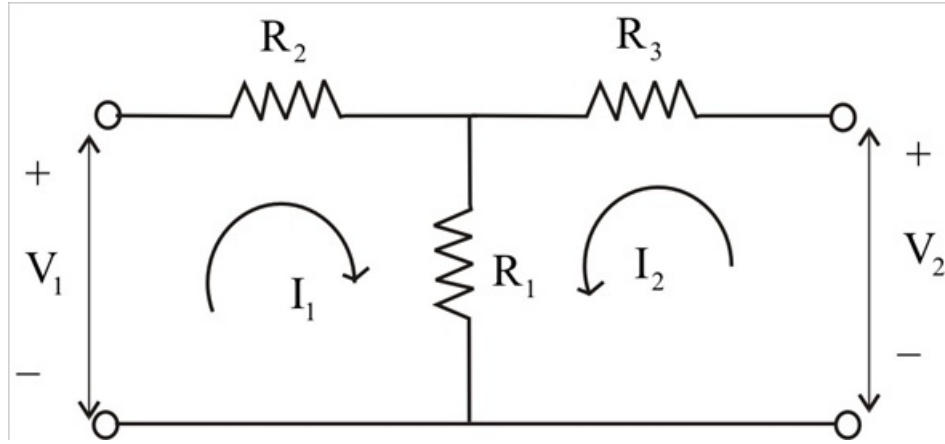
Consider π network which can be converted into T network. Then

Step 3 of 10



To determine the T parameters for the T-network

Consider the circuit



Step 5 of 10

Using mesh analysis we have

$$V_1 = I_1 (R_1 + R_2) + I_2 R_2$$

$$V_2 = I_2 (R_3 + R_1) + I_1 R_2$$

$$\begin{bmatrix} (R_1 + R_2) & R_2 \\ R_2 & (R_2 + R_3) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$\text{so, } [Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} (R_1 + R_2) & R_2 \\ R_2 & (R_2 + R_3) \end{bmatrix}$$

$$\text{so, } \Delta Z = (R_1 + R_2)(R_2 + R_3) - R_2^2 = R_1 R_2 + R_1 R_3 + R_2 R_3$$

Step 6 of 10

$$[T] = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \frac{1}{Z_{21}} \begin{bmatrix} Z_{11} & \Delta Z \\ 1 & Z_{22} \end{bmatrix}$$

$$[T] = \frac{1}{R_2} \begin{bmatrix} (R_1 + R_2) & (R_1 R_2 + R_1 R_3 + R_2 R_3) \\ 1 & (R_2 + R_3) \end{bmatrix}$$

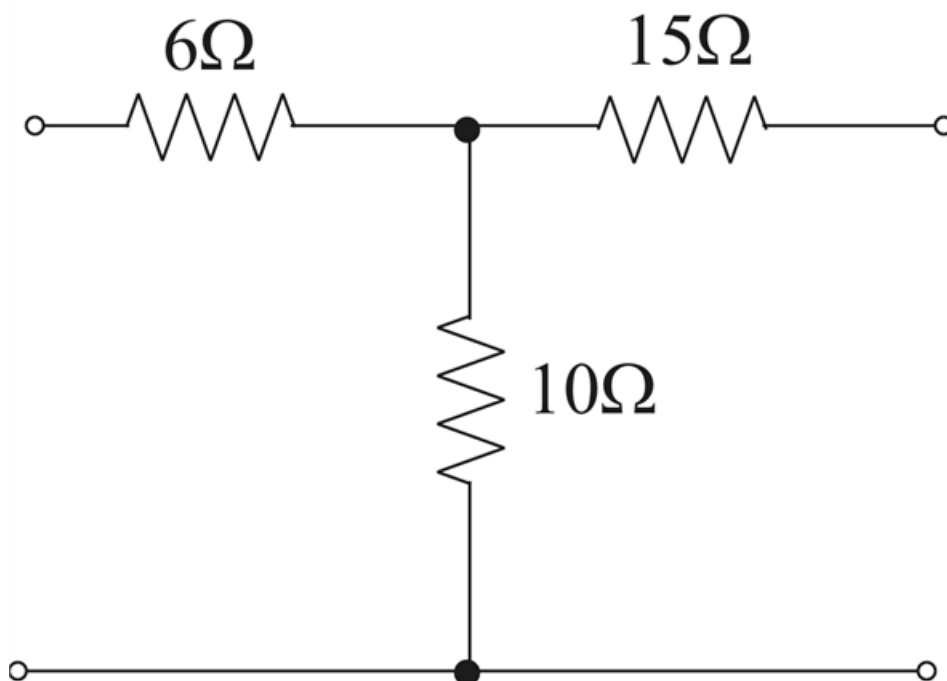
$$A = \frac{R_1 + R_2}{R_2} = 1 + \frac{R_1}{R_2}$$

$$B = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2} = R_3 + \frac{R_1}{R_2} (R_2 + R_3)$$

$$C = \frac{1}{R_2}$$

$$D = \frac{R_2 + R_3}{R_2} = 1 + \frac{R_3}{R_2}$$

Step 7 of 10



Step 8 of 10

By substituting in above equations

$$A = 1 + \frac{6}{10}$$

$$A = 1.6$$

$$B = 15 + \frac{6}{10}(10 + 15)$$

$$B = 30$$

$$C = \frac{1}{10}$$

$$C = 0.1$$

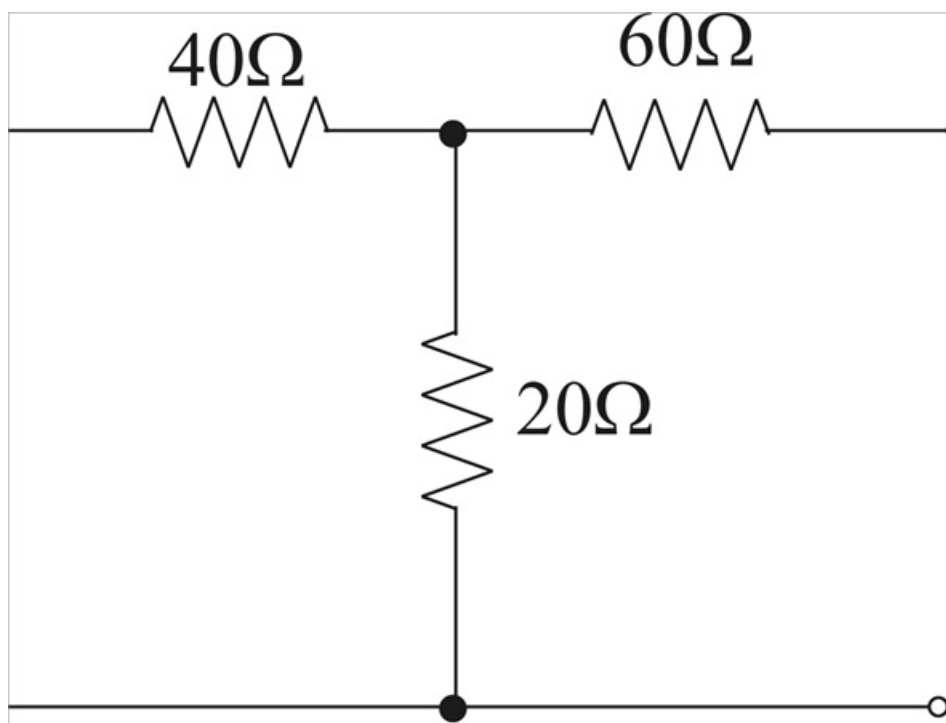
$$D = 1 + \frac{15}{10}$$

$$D = 2.5$$

And in matrix form is

$$[T_a] = \begin{bmatrix} 1.6 & 30\Omega \\ 0.1S & 2.5 \end{bmatrix}$$

Step 9 of 10



Step 10 of 10

$$A = 1 + \frac{40}{20}$$

$$A = 3$$

$$B = 60 + \frac{40}{20}(20 + 60)$$

$$B = 220$$

$$C = \frac{1}{20}$$

$$C = 0.05$$

$$D = 1 + \frac{60}{20}$$

$$D = 4$$

Therefore, the transmission parameter in matrix form is

$$[T_b] = \begin{bmatrix} 3 & 220\Omega \\ 0.05\text{S} & 4 \end{bmatrix}$$

Thus, for the total network

$$[T] = [T_a][T_b]$$

$$[T] = \begin{bmatrix} 1.6 & 30 \\ 0.1 & 2.5 \end{bmatrix} \begin{bmatrix} 3 & 220 \\ 0.05 & 4 \end{bmatrix}$$

$$[T] = \begin{bmatrix} 1.6 \times 3 + 30 \times 0.05 & 1.6 \times 220 + 30 \times 4 \\ 0.1 \times 3 + 2.5 \times 0.05 & 0.1 \times 220 + 2.5 \times 4 \end{bmatrix}$$

$$[T] = \begin{bmatrix} 6.3 & 472\Omega \\ 0.425\text{S} & 32 \end{bmatrix}$$